

1. Name of the medicinal product

DIACOL

2. Qualitative and Quantitative Composition

Contains 27 mg of dextromethorphan hydrobromide per 15 ml of syrup. Excipients, see 6.1

3. Pharmaceutical form

Syrup.

4. Clinical Particulars**4.1 Therapeutic Indications**

For the temporary relief of dry coughs due to irritation of the upper or lower respiratory tract.

4.2 Posology and method of administration

The following dose scheme is suggested:

Adolescents, the elderly and adults: 15 ml every 6 to 8 hours. Maximum daily dose of 60 ml.

Children between 2 and 12 years of age: dose according to doctor instructions. Between 6 and 12 years of age the maximum daily dose is 30 ml, and between 2 and 6 years of age the maximum daily dose is 15 ml.

Children under 2: not recommended for use in this age group.

DIACOL may be administered every 4 hours, reducing the amount of syrup per take so as not to exceed the maximum daily dosage.

Do not exceed the recommended doses or use continuously for extended periods of time.

4.3 Contraindications

Hypersensitivity to any of the components of DIACOL. Productive cough (with expectoration). Concomitant treatment with monoamine oxidase inhibitors.

4.4 Special warnings and precautions for use

DIACOL should be used with caution in atopic or asthmatic patients, as it may trigger the release of histamine.

Use with caution in sedated or frail patients.

Patients with liver impairment may suffer changes in metabolism.

Cases of dextromethorphan abuse and dependence have been reported. Caution is particularly recommended for adolescents and young adults as well as in patients with a history of drug abuse or psychoactive substances.

If cough persists for longer than 1 week, recurs or is accompanied by high fever, skin rash or intense headache the patient should seek a doctor.

DIACOL contains 121.64 g of sucrose per 200 ml of syrup. Each 15 ml dose contains approximately 9.1 g of sucrose. Should not be used by patients with hereditary fructose intolerance, glucose-galactose malabsorption syndrome or sucrase-isomaltase deficiency. DIACOL contains 2.75 g of 96% ethanol per 200 ml of syrup. Each 15 ml dose contains approximately 0.20 g of 96% ethanol. It may be harmful to patients who suffer from alcoholism, liver disease, epilepsy, or brain disease or injury, as well as for pregnant women and children. It may alter or enhance the effects of other medicines.

Dextromethorphan is metabolised by hepatic cytochrome P450 2D6. The activity of this enzyme is genetically determined. About 10% of the general population are poor

metabolisers of CYP2D6. Poor metabolisers and patients with concomitant use of CYP2D6 inhibitors may experience exaggerated and/or prolonged effects of dextromethorphan. Caution should therefore be exercised in patients who are slow metabolizers of CYP2D6 or use CYP2D6 inhibitors (see also section 4.5).

Serotonin Syndrome

Serotonergic effects, including the development of a potentially life-threatening serotonin syndrome, have been reported for dextromethorphan with concomitant administration of serotonergic agents, such as selective serotonin re-uptake inhibitors (SSRIs), drugs which impair metabolism of serotonin (including monoamine oxidase inhibitors (MAOIs)) and CYP2D6 inhibitors.

Serotonin syndrome may include mental-status changes, autonomic instability, neuromuscular abnormalities, and/or gastrointestinal symptoms.

If serotonin syndrome is suspected, treatment with Diacol should be discontinued.

Paediatric population

Serious adverse events may occur in children in case of overdose including neurological disorders. Caregivers should be advised not to exceed the recommended dose.

4.5 Interaction with other medicinal products and other forms of interaction

Monoamine oxidase (MAO) inhibitors:

The concomitant use of DIACOL and MAO inhibitors can precipitate adrenergic crises, coma, dizziness, excitability, hypertension, hyperpyrexia, intracerebral haemorrhage, lethargy, nausea, psychotic behaviour, muscle spasms, shaking and possibly death, therefore the simultaneous administration of both is contraindicated. A period of 14 days must be allowed between the interruption of MAO inhibitors and the introduction of dextromethorphan.

Central nervous system depressants:

When used concomitantly, dextromethorphan and central nervous system depressants may produce additive effects.

CYP2D6 inhibitors:

Dextromethorphan is metabolized by CYP2D6 and has an extensive first-pass metabolism. Concomitant use of potent CYP2D6 enzyme inhibitors can increase the dextromethorphan concentrations in the body to levels multifold higher than normal. This increases the patient's risk for toxic effects of dextromethorphan (agitation, confusion, tremor, insomnia, diarrhoea and respiratory depression) and development of serotonin syndrome. Potent CYP2D6 enzyme inhibitors include fluoxetine, paroxetine, quinidine and terbinafine. In concomitant use with quinidine, plasma concentrations of dextromethorphan have increased up to 20-fold, which has increased the CNS adverse effects of the agent. Amiodarone, flecainide and propafenone, sertraline, bupropion, methadone, cinacalcet, haloperidol, perphenazine and thioridazine also have similar effects on the metabolism of dextromethorphan. If concomitant use of CYP2D6 inhibitors and dextromethorphan is necessary, the patient should be monitored and the dextromethorphan dose may need to be reduced.

4.6 Pregnancy and lactation

Although there are no documented foetal problems secondary to the use of DIACOL in pregnancy, or any data revealing any type of interference with the reproductive system, DIACOL is not recommended during pregnancy.

Likewise DIACOL should not be used during lactation.

4.7 Effects on the ability to drive and use machines

No known interferences at the recommended therapeutic doses.

4.8 Undesirable effects

These are rare, there may be mild disturbances of the gastrointestinal tract such as nausea, vomiting and epigastric pain, dizziness and vertigo.

4.9 Overdose

Symptoms and signs:

Dextromethorphan overdose may be associated with nausea, vomiting, dystonia, agitation, confusion, somnolence, stupor, nystagmus, cardiotoxicity (tachycardia, abnormal ECG including QTc prolongation), ataxia, urinary retention, toxic psychosis with visual and auditory hallucinations, hyperexcitability.

In the event of massive overdose the following symptoms may be observed: coma, respiratory depression, convulsions.

Management:

-Activated charcoal can be administered to asymptomatic patients who have ingested overdoses of dextromethorphan within the preceding hour.

-For patients who have ingested dextromethorphan and are sedated or comatose, naloxone, in the usual doses for treatment of opioid overdose, can be considered. Benzodiazepines for seizures and benzodiazepines and external cooling measures for hyperthermia from serotonin syndrome can be used.”

5. Pharmacological properties**5.1 Pharmacodynamic properties**

Pharmacotherapeutic group: VI-4-b - Fluidizing agent, antitussive and expectorant: other antitussives *ATC code:* R05D A09.

Dextromethorphan is a centrally acting antitussive. It is an isomer of levorphanol, a codeine analogue. Unlike the other isomers, dextromethorphan has no analgesic properties, is not habit forming and does not act on the opioid receptors. Its antitussive properties are the result of its direct action on the cough centre in the medulla, raising for the threshold of the cough reflex. Its antitussive action is similar to that of codeine and its effects last for 6 hours. Unlike codeine, its sedative effect is essentially negligible and observed only at extremely high doses. At therapeutic doses it does not inhibit the ciliary activity of the respiratory tract, which is an advantage compared to other antitussive drugs.

5.2 Pharmacokinetic properties

Dextromethorphan is rapidly absorbed in the gastro-intestinal tract. Its t_{max} is 2 to 2.5 hours and C_{max} following a single 60 mg dose is 5.2 ng/mL. The onset of its antitussive effect is 15 to 30 minutes after administration, and lasts for 5 to 6 hours.

Dextromethorphan distributes in the CNS, reaching detectable concentrations in the cerebrospinal fluid (CSF). In animal studies CSF/plasma concentration ratios varied between 32.8 and 80%.

Dextromethorphan undergoes rapid and extensive first-pass metabolism in the liver after oral administration. Genetically controlled O-demethylation (CYD2D6) is the main determinant of dextromethorphan pharmacokinetics in human volunteers.

It appears that there are distinct phenotypes for this oxidation process resulting in highly variable pharmacokinetics between subjects. Unmetabolised dextromethorphan, together with the three demethylated morphinan metabolites dextrophan (also known as 3-hydroxy-N-methylmorphinan), 3- hydroxymorphinan and 3-methoxymorphinan have been identified as conjugated products in the urine.

Dextrophan, which also has antitussive action, is the main metabolite. In some individuals metabolism proceeds more slowly and unchanged dextromethorphan predominates in the blood and urine.

Dextromethorphan and its demethylated metabolites are excreted in the urine. The elimination half-life of dextromethorphan is 1.4 to 3.9 hours, and of dextrophan is 3.5 to 5.6 hours.

5.3 Pre-clinical safety data

Dextromethorphan is a drug in current use and included in almost all pharmacopoeias.

6. Pharmaceutical Particulars

6.1 List of excipients

Sucrose, anhydrous citric acid, methylparaben, liquid essence of orange, 96 percent ethanol and purified water.

6.2 Incompatibilities

In the absence of incompatibility studies, this medicinal product should not be mixed with other medicinal products.

6.3. Shelf life

5 years.

After first opening: 4 years.

Do not use this medicine after the expiry date, which is stated on the bottle and the carton after EXP. The expiry date refers to the last day of that month.

6.4. Special precautions for storage

No special precautions for storage.

6.5 Nature and content of the container

Cardboard carton containing one type III amber glass bottle and a package leaflet. The packaging includes a dosing cup. The bottle contains 200 ml of syrup.

6.6. Special precautions for use and handling (and disposal)

No special requirements

7. Marketing Authorization Holder

BIAL - Portela & C^a, S.A.

À Avenida da Siderurgia Nacional

4745-457 S. Mamede do Coronado

8. Marketing Authorization Numbers

2008290

9. Date of Renewal of Marketing Authorization

10. Date of (partial) revision of the text

10/2019